PW SKILLS

DATA PREPARATION IN EXCEL



1. What is data cleaning, and why is it important in data analysis?

Perform necessary actions to:

- What are the potential consequences of analyzing unclean or messy data?
- Explain the common steps involved in cleaning and organizing data.

2. Apply Data Validation to restrict Quantity values to only whole numbers between 1 and 10.

a. Configure an input message that appears when a user selects a cell in the "Quantity" column explaining:

"Please enter a whole number between 1 and 10."

b. Set up an error alert message that triggers if the user enters a number less than 1 or greater than 10, showing:

"Invalid input! The quantity must be a whole number between 1 and 10."

Write a step-by-step approach for this question

Customer Name	Product Name	Category	Quantity	Unit Price ($)
Jane Smith	Shoes	Electronics		81
Isabella Moore	Laptop	Electronics		121
Daniel Davis	Sofa	Clothing		239
Alex Moore	Shoes	Electronics		500
Michael Johnson	Table Lamp	Home Decor		423
Daniel Johnson	Backpack	Electronics		160
Isabella Davis	Headphones	Electronics		348
Jane Davis	Headphones	Electronics		152
Alex Wilson	T-shirt	Home Decor		369

3. What is the difference between data validation and data cleaning?

4. You are provided with a raw sales dataset collected from multiple sources. Before performing any business analysis, ensure that the dataset is accurate, consistent, and reliable.

Transform the dataset so that it can be confidently used for reporting by addressing:

- Data inconsistencies across records
- Explain the common steps involved in cleaning and organizing data.
- Invalid or illogical values
- Formatting and standardization issues
- Missing or incorrect data

Dataset: Link

5. In a business environment, data processing tasks are often repeated on a daily basis.

Explain when macros should be used in Excel.

Also, identify situations where using macros may NOT be appropriate.

6. You are analyzing sales performance data for a business. Before presenting insights, enhance the dataset visually so that key performance indicators can be **quickly identified and interpreted.**

Apply appropriate techniques to:

- Identify products that are **underperforming against targets**
- Highlight records with **negative profitability**
- Distinguish between **achieved, exceeded, and missed targets**
- Make the dataset **visually intuitive for decision-making**

Dataset: Link

7. You are provided with a dataset containing transaction details.
Before using this data for financial analysis, verify whether the calculations are **accurate and reliable.**

Perform necessary actions to:

- Identify records where the **Total Amount is incorrect**
- Ensure that all calculations follow the correct business logic
- Highlight or correct discrepancies in the dataset
- Make the dataset consistent and reliable for reporting

Dataset: Link

8. You are provided with a dataset where multiple pieces of information are stored in single columns.
Before performing any analysis, restructure the dataset into a **well-organized and usable format.**

Transform the dataset so that:

- Customer details are separated into **individual components**
- Product information is structured into **distinct columns**
- Order details are converted into **usable numerical fields**
- The dataset follows a **tabular format suitable for analysis**

Dataset: Link

9. You are provided with a dataset containing incomplete transaction details.
Before using this data for analysis, determine how missing values should be handled to ensure accuracy and reliability.

Identify missing values across columns

- Identify records where the **Total Amount is incorrect**
- Decide appropriate treatment for each column based on its business meaning
- Ensure that the dataset remains **logically consistent after handling missing data**
- Avoid introducing incorrect assumptions while filling missing values

Dataset: Link

10. You are working with customer transaction data collected from multiple sources.
Before analysis, ensure that duplicate and near-duplicate records do not distort results.

Perform necessary actions to:

- Identify **exact duplicate records**
- Detect **near-duplicate entries** caused by variations in naming (e.g., case differences, abbreviations)
- Ensure that each unique customer transaction is represented only once
- Maintain data accuracy without losing valid records

Dataset: Link

11. You are given a dataset where calculations are resulting in errors due to inconsistent and invalid inputs.
Before performing any financial analysis, ensure that all calculations are handled in a way that prevents errors and maintains data reliability.

Perform necessary actions to:

- Identify records where calculations are failing
- Ensure formulas handle invalid or missing inputs gracefully
- Prevent errors from appearing in the final dataset
- Avoid introducing incorrect assumptions while filling missing values

Dataset: Link

12. You receive a similar dataset every day, but the number of rows keeps changing.
Each time, you are required to prepare the data in a specific format before sharing it with your manager.

However, when trying to automate this process, you notice that the solution does not work correctly for datasets of different sizes.

Design an approach that:

- Automates the process efficiently
- Works correctly even when the dataset size changes
- Avoids errors caused by fixed or static steps
- Ensures consistent output across multiple datasets

Dataset: Link